Objectives

The main objective of the project is :

1. Maintain the temperature of the green house using temperature sensor.
2. Automatically watering the plants in the greenhouse as per the soil gets dry .
3. Provide security using RFID sensor for the locker.
4. Air quality checking for the betterment of the crops.
5. Provide user interface for the farmer with android app and web application.
6. Maintaining the database of the previous day's details and conditions of the green house.

Hardware Requirements:

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Monitor : 15 VGA Colour.
- Mouse : Logitech.
- Ram : 512 Mb.

Software Requirements:

- Operating system : - Windows XP.
- Coding Language : -JAVA,JSP, HTML, Arduino, Processing.
- Tool Used : Dreamviewer.
- Server :Tomcat
- Database Server : Mysql.

IOTs

- Arduino board
- Temperature and humidity sensor
- Moisture sensor
- RFID sensor
- Servo motor
- Water pump motor
- Air quality checking sensor

Methodology

As per the temperature is need the system will cool down the temperature of the green house by water sprinkler. If the temperature of the house is low then it will power on the heater to worm the house.

As per the soil level the system will switch on the motor to water the plants. If the moisture level is high then it will be switched off automatically.

For the security protection we have applied RFID sensors to identify the user and open the doors for the users.

We also monitor the air quality of the green house and indicate the level to the user regarding the quality level of the air. This helps the farmer to maintain the good air quality in the green house.

All the status of the green house data are stored in the database and shows the status to the user about the conditions of the green house.